## Homework

**Homework 12 — Projection and intersection — 55 min.**

In an orthonormal frame of space, consider the point $M(3;0;0)$ and the planes $\mathcal{P} : x + 2y + 2z = 9$ and $\mathcal{Q} : x - y = 0$.

1. Give the coordinates of a non-zero normal vector to $\mathcal{P}$.
2. Determine the coordinates of the orthogonal projection $H$ of $M$ onto $\mathcal{P}$, then calculate the distance from $M$ to $\mathcal{P}$.
3. Show that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it.
4. Does the point $H$ belong to this line? Justify.
5. Determine the coordinates of the point $K$ of this line whose third coordinate (the "cote") is zero. Calculate $MK$, then compare $MK$ and $MH$.